Mid Semester Examination March 2025

Name of the Program: B.Tech. Civil Engineering — *Semester:* VI

Name of the Course: Water Resources Engineering II — *Course Code:* TCE-603

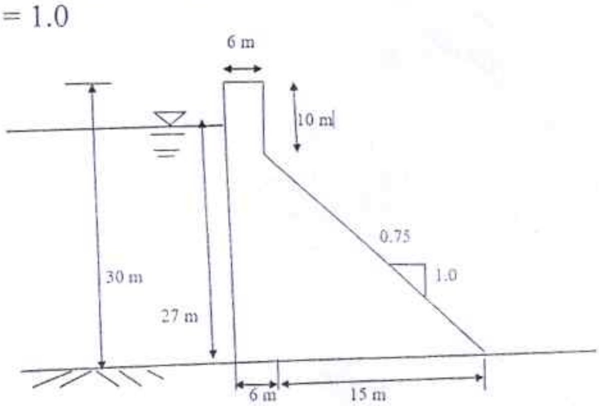
Time: $1\frac{1}{2}$Hour — Maximum Marks: 50

Note:

(i) Answer all the questions by choosing *any one of the sub questions.*

(ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	Enumerate different methods of stability analysis of gravity dam and explain in brief any one of them.	CO-1
	OR	
(b)	Design an elementary profile of a gravity dam to store water up to a depth of 40 m. assuming following data. Uplift factor =0.75 Specific gravity of dam material = 2.4 Coefficient of friction = 0.7	
Q2	(10 marks)	
(a)	Explain briefly with neat sketch's the different forces that may act on a gravity dam.	CO-1
	OR	
(b)	Determine the stability analysis of a gravity dam with the following data. Overturning moment at toe = 1×10^6KN.m Total resisting moment at toe = 2×10^6KN.m Total vertical force above base = 50,000 KN Base width of the dam = 50 m. Slope of the downstream surface = 0.8 H : 1 V. Calculate the maximum and minimum vertical stresses on the foundation and also determine the maximum principal stress at the toe of the dam. Assume that there is no tail water and u/s face is vertical.	
Q3	(10 marks)	
(a)	Draw a neat sketch (section) of an earthen dam and name the various parts.	CO-2
	OR	
(b)	Determine the phreatic line for an earthen dam homogeneous in section from the following data: Height of dam = 20 m. Free board = 2.0 m. Coefficient of permeability of material of the dam = 4×10^{-4} cm/sec Top width = 4 m. Upstream slope = 3:1 Downstream slope = 3:1 And also find discharge per meter length of the earthen dam	

Q4	(10 marks)	
(a)	Prove that the most economical section of an arch dam has a central angle of $133^0 34'$.	CO-2
	OR	
(b)	Derive relations to determine seepage discharge in an earth dam and calculate the discharge per meter length of the earthen dam from the following data: Height of dam = 24 m. Free board = 2.0 m. Coefficient of permeability of material of the dam = 4×10^{-4} cm/sec From the flow net for the dam (homogeneous in section) the following information was available: Total number of flow channels = 6 Total number of potential drops = 14	
Q5	(10 marks)	
(a)	For the gravity dam shown in fig Determine-i). Factor of safety against overturning ii). Factor of safety against sliding Considering Water pressure, weight of dam and uplift pressure only. Given $\mu = 0.7, \gamma_c = 24$ KN/m^3, $\gamma_w = 10$ KN/m^3 and uplift pr. Coefficient c = 1.0  	CO-1 & CO-2
	OR	
(b)	To determine the factor of safety of the d/s slope during steady seepage, the section of a homogeneous earth dam was drawn to scale of 1 cm = 10 m ; and the following results were obtained on a trial slip circle. Area of N-diagram= 12.15 sq cm. Area of T-diagram= 6.50 sq cm Area of U-diagram= 4.02 sq cm. Length of arc= 11.60 cm. The dam material has the following properties: Effective angle of internal friction = 26° Cohesion = 19.5 kN/m2 Unit weight of soil = 19.0 kN/m3 Determine the factor of safety of the slope.	